

1번

```
A1 = [1 4 2; 2 4 100; 7 9 7; 3 pi 42];  
B1 = log(A1);  
  
row2_B = B1(2, :)
```

```
row2_B = 1×3  
    0.6931    1.3863    4.6052
```

```
row2_B_sum = sum(row2_B)
```

```
row2_B_sum =  
    6.6846
```

```
col_mul = B1(:, 2) .* A1(:, 1)
```

```
col_mul = 4×1  
    1.3863  
    2.7726  
   15.3806  
    3.4342
```

```
max_vec = max(col_mul)
```

```
max_vec =  
   15.3806
```

```
div_vec = A1(1, :) ./ B1(1:3,3);  
div_vec_sum = sum(div_vec)
```

```
div_vec_sum = 1×3  
    2.1737    8.6950    4.3475
```

2번

```
v02 = 50;  
A2 = deg2rad(50);  
g2 = 9.81;  
  
thit = (2*v02*sin(A2)) / g2
```

```
thit =  
    7.8088
```

```
ymax = (v02 * sin(A2))^2 / (2 * g2)
```

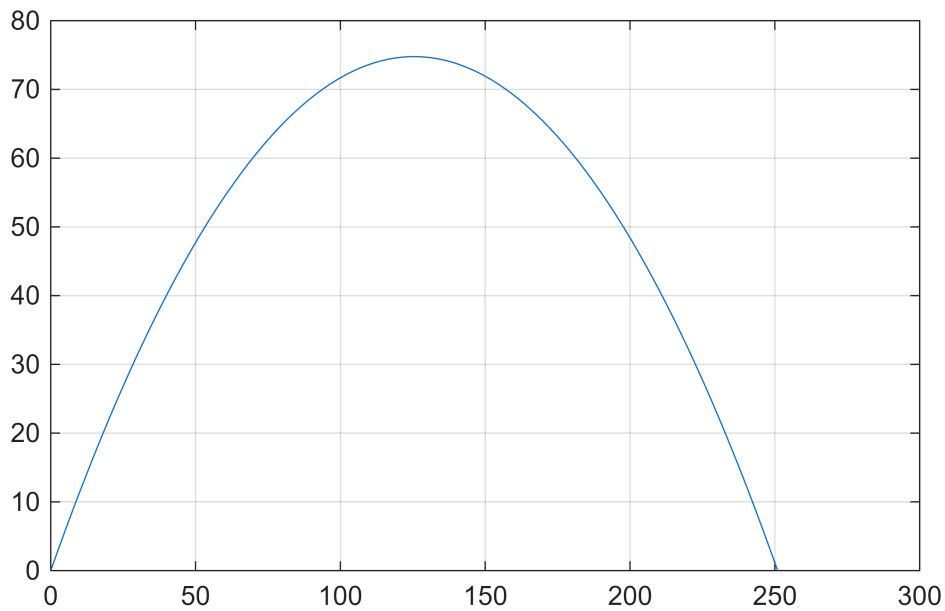
```
ymax =  
   74.7737
```

```
t2 = linspace(0, thit, 100);  
x2 = (v02 * cos(A2)) .* t2;
```

```

y2 = (v02 * sin(A2)) .* t2 - 0.5 * g2 * t2.^2;
plot(x2, y2)
grid on

```



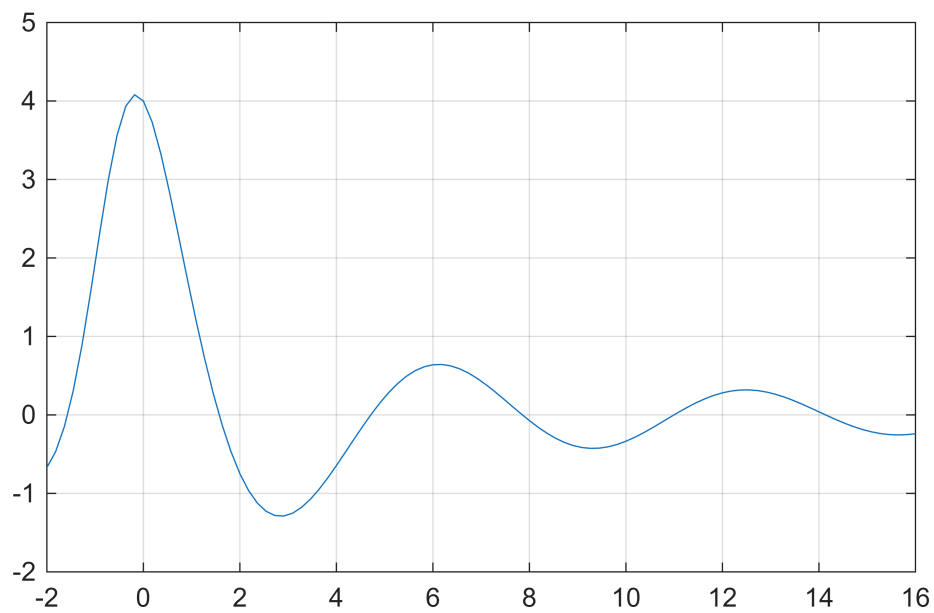
3번

```

x3 = linspace(-2,16);
f3 = (4*cos(x3)) ./ (x3+exp(-0.75*x3));

figure
plot(x3, f3)
grid on

```



4번

```
C4 = [5 4 6; 3 2 4; 6 5 3; 3 5 4; 2 4 3]
```

```
C4 = 5×3
      5     4     6
      3     2     4
      6     5     3
      3     5     4
      2     4     3
```

```
prices = [300; 550; 400; 250; 500];
```

```
monthly_total = sum(C4,1)
```

```
monthly_total = 1×3
      19      20      20
```

```
material_total = sum(C4,2)
```

```
material_total = 5×1
      15
       9
      14
      12
       9
```

```
all_total = sum(C4, "all")
```

```
all_total =
      59
```

5번

```
p5_1 = [5,0,-3,7,10];
p5_2 = [4,-7,9,2];
ans_mul = conv(p5_1, p5_2)
```

```
ans_mul = 1×8
      20     -35      33      59     -36     -13     104      20
```

```
p5_3 = [5,-7,2,8];
p5_4 = [6,3,-2];
[q5, r5] = deconv(p5_3, p5_4)
```

```
q5 = 1×2
      0.8333     -1.5833
r5 = 1×4
      0           0      8.4167      4.8333
```

```
p5_5 = [36,-8,0,2];
p5_6 = [36,-5,4,-3];
val_x5 = polyval(p5_5, 5) / polyval(p5_6, 5)
```

```
val_x5 =  
0.9795
```

7번

```
bridges(1) = struct('Location', 'Smith St.', 'MaxLoad', 80, 'YearBuilt', 1928,  
'DueYear', 2025);  
bridges(2) = struct('Location', 'Hope Ave.', 'MaxLoad', 90, 'YearBuilt', 1950,  
'DueYear', 2027);  
bridges(3) = struct('Location', 'Clark St.', 'MaxLoad', 85, 'YearBuilt', 1933,  
'DueYear', 2024);  
bridges(4) = struct('Location', 'North Rd.', 'MaxLoad', 100, 'YearBuilt', 1960,  
'DueYear', 2023);  
bridges
```

bridges = 1×4 struct

필드	Location	MaxLoad	YearBuilt	DueYear
1	'Smith St.'	80	1928	2025
2	'Hope Ave.'	90	1950	2027
3	'Clark St.'	85	1933	2024
4	'North Rd.'	100	1960	2023

```
bridges(3).DueYear = 2026
```

bridges = 1×4 struct

필드	Location	MaxLoad	YearBuilt	DueYear
1	'Smith St.'	80	1928	2025
2	'Hope Ave.'	90	1950	2027
3	'Clark St.'	85	1933	2026
4	'North Rd.'	100	1960	2023

```
bridges(5).Location = 'Shore Rd.';  
bridges(5).MaxLoad = 85;  
bridges(5).YearBuilt = 1997;  
bridges(5).DueYear = 2022;  
bridges
```

bridges = 1×5 struct

필드	Location	MaxLoad	YearBuilt	DueYear
1	'Smith St.'	80	1928	2025
2	'Hope Ave.'	90	1950	2027
3	'Clark St.'	85	1933	2026
4	'North Rd.'	100	1960	2023

필드	Location	MaxLoad	YearBuilt	DueYear
5	'Shore Rd.'	85	1997	2022